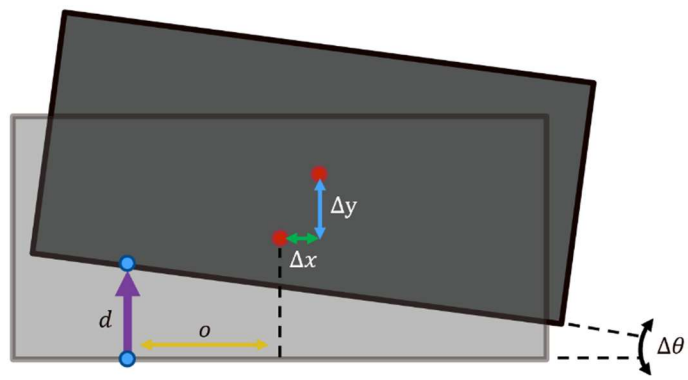
Project 2 Physics Equation Clarification

For physics estimation of the pushing task, we want to calculate the Δx, Δy, and $\Delta\theta$ based on the push parameters, o (offset) and d (distance). Here, we assume the side being pushed has already been decided. The velocity profile is designed such that the total travel distance equals d. Therefore, the blue point on the object where the force is being applied, travels for d approximately (in reality, this point moves along a curve, not a straight line). As the object rotates at the same time, this distance is divided along the x and y directions. To calculate the final Δx and Δy, we can discretize the entire velocity trajectory, calculate the change at each smaller timestep Δt, and then sum them over the total time $T = 3s$. By using these equations, we take the actual rotation angle into account:

$$\Delta x = \sum_{i=0}^{N} V(i)\cos\big(\theta(i)\big)\,\Delta t,$$

$$\Delta x = \sum_{i=0}^{N} V(i)\sin\big(\theta(i)\big)\,\Delta t.$$



The angle profile can be obtained from the equation below:

$$\Delta\theta = \sum_{i=0}^{N} o.V(i)\,\Delta t,$$

where o is the offset annotated in the figure and is the second push parameters in your dataset. Essentially, you will need a loop to calculate each of these values iteratively, as the rotation changes at each step. You may also use tensor operations as well if you prefer.

Bonus Credits for Project 2

You are encouraged to be creative with your neural network's architecture design. You may also change the physics estimation to find a more accurate model, as long as it's correct and is clearly explained in your report. At the end, your hybrid model is expected to perform and the others. After the validation process, a lower loss indicates a more optimized model. The team with lowest final validation losses (based on your configuration specified in your YAML file) will receive 10 bonus credits.